R code used for figures

Graphical Abstract – PNG designed by Jason Guinan

Figure 1 – PNG designed by Jason Guinan

Figure 2 – PW-DNA_Map_Final.R (with post edits)

Figure 3 - Taxon_sorted_heatmap_with_taxon_total_count.R (with post edits)

Figure 4 – Geochem_Barplot.R

Figure 5 – Bubble_Plot_Methanogens_pdf_save.R (with post edits)

Figure 6 – venn_diagram_sorted.R

Figure S1 - Taxon_IMG_heatmap.R

Figure S2 - Methanogens_MAGs_barplot.R

Figure S3 - Bubble_plot_primer_grouping_basin_width_ajusted.R

How each figure was made:

Graphical Abstract and Figure 1: Jason Guinan

Figure 2: Need two file types:

- One that has Well.Type, Molecular.Type, longitude and latitude file names.
- The ene_014_shale_basins/playsAndBasins/Data_02142014.gdb
- Run the R code: PW-DNA_Map_Final.R

Figure 3:

- Filter out the genes of interest in the narrative for genomes/MAGs (https://narrative.kbase.us/narrative/264125) for and extract them as a .csv.
- Then run the python code and command:
    - python merge_gene_hit_files.py -i gene_table/*.csv -o merged_thiosulfate_genes.tsv
- Run the R code: Taxon_sorted_heatmap_with_taxon_total_count.R

Figure 4: Need one file:

- Generate file in PW-DNA Database Metadata Geochem excel file in the column titled Tables

- One column list of metadata or geochem condition other column percentage of filled
- Run R code: Geochem_Barplot.R

Figure 5:

- Filter out all North America samples by country
- Follow filter steps to get to select samples
- Filter by methanogens
- Generate a subset of samples
- Create taxonomy file
- Use file North_america_amplicon.txt
- Use R code: Bubble_Plot_Methanogens_pdf_save.R

Figure 6:

- Generate file that looks like: Figure_6_update_v2.txt
- Run R code: venn_diagram_sorted.R

Figure S1

- Filter genomes by keywords: coal, shale, oil reservoir, produced water, fossil fuel
- Filter out genes of interest in functions for COG and KEGG orthology: dsrA, dsrB, phsA, and aprA
- Run Analyze > Functions > Relative Abundance
- Compare Genome Set to Function Set and produce a Gene Count Heat Map
- Need to make sure taxon information from IMG GTDB columns are marked before download
- Extract them and format the file to look like IMG_thiosulfate_all.tsv
- Run R code: Taxon_IMG_heatmap.R

Figure S2:

- Filter out all methanogen MAGs on database
- Filter out so only Canada and USA
- Generate file that looks like: supplemental_figure1_udate.csv
- Run R code: Methanogens_MAGs_barplot.R

Figure S3

- Use a file that looks like:North_America_amplicon_by_primer
- Run R code: Bubble_plot_primer_grouping_basin_width_ajusted.R
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